INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

SENSOR, GRID WIRE, DT-545/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the grid wire sensor and an installation tool used to install the grid wire. The grid wire sensor is a break wire sensing device used to detect forced entry through a wall, floor, ceiling or door into an area which has been secured. It is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir, Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350

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MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-2000	-	Standard Requirements for Soldered Electrical & Electronic Assemblies.
MIL-STD-45662	-	Calibration System Requirements.

(Unless otherwise specified, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13220E3401	-	Tool, Grid Wire Installation.
TA13220E3402	-	Sensor Assembly, Grid Wire.
C13220E3406	-	Wire, Grid.
TA13220E3436	-	Sensor, Grid Wire.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained form the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JIC, Fort Belvoir, VA 22060-5606.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

D9.1 - Welding of Sheet Metal.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The grid wire sensor (hereinafter called "GWS") shall be in accordance with top assembly TA13220E3436 and as specified herein. The grid wire installation tool (hereinafter called "ITL") shall be in accordance with top assembly TA13220E3401 and as specified herein.

3.1.1 Major components of GWS. The GWS shall consist of one each of the following major components:

a. Gridwire sensor assembly - TA13220E3402 (see 3.4.1, 3.4.2 and 3.4.3).
b. Wire, grid - C13220E3406 (see 3.4.5).

When specified (see 6.2), either of the major components listed above shall be furnished separately.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall

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provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the GWS and ITL. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The GWS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the GWS may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the GWS may be newly fabricated from recovered materials to the maximum extent practicable, provided the GWS produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the GWS.

3.4 Detailed technical requirements. The GWS shall perform as specified herein under any of the operating test conditions specified herein.

3.4.1 Tamper switch operation. An internal tamper switch shall be provided within the GWS assembly which is activated (output contacts closed) when the cover is in place and shall open when the cover is moved not less than 0.125 inch or more than 0.250 inch from the normally closed position and before the cover flanges disengage from the chassis. A pry-off tamper switch shall be provided, the output contacts of which shall be closed when the GWS assembly is mounted on a flat surface and shall open upon raising of the chassis not less than 0.125 inch nor more than 0.250 inch from its mounted position.

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3.4.2 GWS terminal board connections. The GWS terminal board (TB1) shall be marked with identifying terminal numbers as follows:

TB1	Function
1	Horizontal grid
2	Horizontal grid
3	Vertical grid
4	Vertical grid
5	Tamper
6	Tamper return
7	Enclosure ground

3.4.3 Alarm conditions. The alarm output conditions across the sensor tamper alarm contacts TB1-5 and TB1-6 shall be as follows:

a. No alarm: less than 1 ohm.
b. Alarm: more than 100 kilohms (k ohms).

3.4.4 ITL. The ITL shall automatically dispense the grid wire and be manually operated to fasten the wire to the foundation board with staple-type fasteners. The ITL shall have tension adjustment such that with proper adjustment the fasteners shall penetrate the mounting surface to a depth of 0.25 $\pm$0.07 inch or as necessary to hold the grid wire firmly without breaking or crimping it. The ITL shall have a magazine capable of dispensing a minimum of 125 fasteners before requiring a refill. The ITL shall be manually refilled by inserting the clips of fasteners into the magazine. The ITL shall develop a tension of not less than 1 pound nor greater than 4 pounds on the wire during application. The wire shall be dispensed through a positive guide system such that no alignment is necessary to assure that the staple will captivate the wire. On sheetrock walls, the staple shall not cut into the wire. The ITL shall be capable of dispensing, a minimum of 600 feet of grid wire. The weight of the ITL, including wire and fasteners, shall not be greater than 4 pounds.

3.4.5 Grid wire. The grid wire shall be 26 AWG enamel-coated, solid copper wire. When a gradually increasing load is slowly applied to the wire, the wire shall break at a load of 8 $\pm$1 pounds. The grid wire shall be wound on a spool containing at least 600 feet of grid wire.

3.5 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts, except for adjustment, and without selection for fit or performance.

3.6 Identification marking. The GWS, ITL and major components thereof shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

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3.7 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the GWS normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G as applicable. Color shall be lusterless gray conforming to FED-STD-595, color No. 36440. Nonpermissible galvanic couples (dissimilar metals) as specified in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.8 Workmanship. Workmanship shall be in accordance with the requirements and guidance pararaphs of MIL-STD-454, requirement 9.

3.8.1 Soldering. All soldering shall be performed in accordance with the general requirements of MIL-STD-2000.

3.8.2 Welding. Sheet metal welding shall be in accordance with AWS D9.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The GWS, ITL and major components when furnished separately shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article GWS, ITL and major components when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 GWS's. GWS's shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.3.2.2 ITL's. ITL's shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection.

4.3.2.3 Major components when furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in column 1 of table I, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for test shall be in accordance with MIL-STD-105 and as specified herein.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 GWS's. Each GWS shall be subjected to the test marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.4.3.1.2 ITL's. Each ITL shall be subjected to the test marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.4.3.1.3 Major components when furnished separately. Each GWS assembly when furnished separately shall be subjected to the test marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

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4.4.3.2 Samples.

4.4.3.2.1 GWS's and grid wire when furnished separately. In normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table I, as applicable. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced inspection shall be 3 units. For each sample size, the lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The GWS, ITL and major components when furnished separately shall be examined as specified herein for the following defects (as applicable):

Major

101. Assembly not correct (see 3.1).
102. Components missing or not as specified (see 3.1 and 3.4.2).
103. Any part or component not in accordance with the quality assurance provisions (QAP) shown on the drawings (see 3.1.2).
104. Materials not as specified (see 3.3).
105. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
107. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
108. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the GWS, ITL or major components when furnished separately (see 3.3.2).
109. Interchangeability not as specified (see 3.5).
110. Identification or special marking missing, incomplete or illegible (see 3.6).
111. Treatment and painting not as specified (see 3.7).
112. Workmanship not as specified (see 3.8).
113. Soldering not as specified (see 3.8.1).
114. Welding not as specified (see 3.8.2).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Unless otherwise specified herein, the GWS and ITL shall operate as specified herein without maintenance.

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4.5.2.2 Test schedule.

TABLE I. Test schedule for GWS and major components when furnished separately.

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X 1/	X 1/	-	Functional-GWS assembly	3.4.1, 3.4.3	4.5.2.4.1
X 2/		X 2/	Wire strength - break test	3.4.5	4.5.2.4.3

1/ Required for procurement of GWS and GWS assembly when furnished separately.
2/ Required for procurement of GWS and grid wire when furnished separately.

TABLE II. Test schedule for ITL.

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Installation tool.	3.4.4	4.5.2.4.2

4.5.2.3 Test methods. Test shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Functional.

4.5.2.4.1 Sensor assembly operation. The sensor assembly shall be tested in accordance with appendix test method 1. Nonconformance to 3.4.1 and 3.4.3 shall constitute failure of this test.

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4.5.2.4.2 ITL. The ITL shall be tested in accordance with appendix test method 2. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.4.3 Grid wire. The grid wire shall be subjected to the wire strength-break test in accordance with appendix test method 3. Nonconformance to 3.4.5 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

115. Preservation method for complete GWS's, ITL's and major components when furnished separately not as specified for levels A, B and C (see 5.1.1, 5.1.2 and 5.1.3).
116. Boxed contents not cushioned to prevent movement or damage as specified for level A, B or C (see 5.1.1.1, 5.2.1.1, 5.1.2 and 5.2.3).
117. Consolidation not as specified for level A, B or C (see 5.1.1.1, 5.2.2 and 5.2.3).
118. Shipping containers not as specified for level A, B or C (see 5.2.1.1, 5.2.2 and 5.2.3).
119. Quantities packed together in containers exceed gross weight specified for level A, B or C (see 5.1.1.2, 5.1.1.2.3, 5.2.1.1, 5.2.2 and 5.2.3).
120. Strapping not as specified for level A, B or C (see 5.2.1.1, 5.2.2 and 5.2.3).
121. Marking missing, illegible, incorrect, or incomplete for levels A, B or C (see 5.3 and 5.3.1).
122. Special marking not as specified (see 5.3.1).

5. Packaging.

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each GWS, ITL and major components when furnished separately shall be preserved by one of the following procedures depending upon procurement of GWS's, ITL's or procurement of major components when furnished separately.

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5.1.1.1 Procurement of complete GWS's. Each GWS shall be preserved in accordance with MIL-P-116 as follows: The GWS assembly shall be preserved in accordance with submethod IIc; each spool of the grid wire shall be individually preserved in accordance with method IC-1 or IC-3. The GWS assembly shall then be individually placed in a close-fitting box conforming to PPP-B-636, W5c, or W6c, style optional and cushioned within the box to prevent movement or damage. The GWS assembly and grid wire shall then be placed together in a box and cushioned as specified herein for the grid wire sensor assembly.

5.1.1.2 Procurement of ITL's. Each ITL and box of staples shall be preserved in accordance with MIL-P-116, method IC-2, using the contractor's carton or a box conforming to PPP-B-566, variety 1, style, type, and class optional, or PPP-B-676, variety 1, style, type, and class optional. These components shall then be placed together in a box and cushioned as specified in 5.1.1.1 in quantities not to exceed the weight or size limitations of the box.

5.1.1.2.1 Major components when furnished separately.

5.1.1.2.2 GWS assembly. GWS assemblies, when furnished separately shall be individually preserved in accordance with MIL-P-116, method IIc. Each GWS assembly shall be placed in a close-fitting box conforming to PPP-B-636, W5c, or W6c, and cushioned within the box to prevent movement within the box.

5.1.1.2.3 Grid wire when furnished separately. When the grid wire is furnished separately, each spool shall be individually preserved as specified in 5.1.1.1 and then placed together in a close-fitting box conforming to PPP-B-636, W5c, W6c, or V3c, cushioned as specified in 5.1.1.1, in quantities not to exceed the weight or size limitations of the box.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except method IA-8 shall be used in place of method IIc.

5.1.3 Level C. Each complete GWS, ITL or major component when furnished separately shall be preserved as specified for level B, except the boxes shall be domestic class.

5.2 Packing. Packing shall be level A, level B, or level C as specified (see 6.2).

5.2.1 Level A. The GWS, ITL and major components when furnished separately, preserved as specified in 5.1, shall be packed by the following procedure depending upon procurement of GWS's, ITL's or major components when furnished separately.

5.2.1.1 Procurement of complete GWS's. Complete GWS's shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the weight and size limitations of the box. Cushioning and blocking within the box used to prevent movement and isolate impact shock shall be in accordance with MIL-STD-1186. Closure

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shall be in accordance with the appendix to the box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable, and ASTM D 4675.

5.2.1.2 Procurement of ITL's. ILT's, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1.

5.2.1.3 Procurement of major components when furnished separately. Major components when furnished separately, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete GWS's, ITL's and major components when furnished separately, preserved as specified in 5.1 shall be packed as specified in 5.2.1 for level A, except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Complete GWS's, ITL's or major components when furnished separately, preserved as specified in 5.1 shall be packed in boxes conforming to PPP-B-636 as specified for level B, except the boxes shall be domestic class and strapping need not be zinc-coated.

5.3 Marking. Marking shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.1.

5.3.1 Special Marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The GWS detects penetration through walls, floors, ceilings, doors and other barriers by a "break wire" method. The grid wire is applied to the surface to be protected in a criss cross pattern such that a wire grid of approximately 4 inch squares, covers the entire surface. The wire grid is then covered with paneling or plywood for protection. Any penetration through this barrier in excess of a 4-inch square results in a break in one or more of the grid wires. The sensor does not produce an intrusion alarm by any means except a physical break in the electrical continuity of the circuit. A break in the electrical continuity of the circuit will result in an alarm being transmitted to, and recognized by, a control unit. The GWS is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Nomenclature and quantity of item(s) to be furnished (see 3.1).
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval and the number of units required (see 3.2).
f. Treatment and painting if other than as specified (see 3.7).
g. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Alarm signals.

6.4.1.1 Intrusion alarm. Warning signal (alarm), presented as a high impedance at the appropriate output terminals, resulting from breaking the grid wire during an intrusion or attempted intrusion into the protected area.

6.4.1.2 Tamper alarm. Warning signal (alarm), presented as a high impedance at the appropriate output terminals, resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the item; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

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6.4.3 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.7 Subject term (key word) listing.

Alarm system
Breakwire sensor
Intrusion detection
Physical security

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activity:
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Army - ME

Project 6350-0116

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APPENDIX

TEST METHODS

FOR

SENSOR, GRID WIRE

AND

INSTALLATION TOOL, GRID WIRE

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether sensors and tools procured under this specification conform to the requirements specified.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

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APPENDIX

TEST METHOD 1

FUNCTIONAL TEST

GRID WIRE SENSOR ASSEMBLY

1. TEST APPARATUS:

   a. Ohmmeter - Simpson 260, or equivalent.
   b. Ruler.

2. TEST PROCEDURE:

   a. Remove junction box assembly cover and connect ohmmeter between TB-5 and TB-6.

   NOTE: Insert ohmmeter leads through conduit entry port to facilitate installation of sensor assembly cover.

   b. Place junction box assembly on a work bench with its bottom or mounting surface down and securely place cover on sensor assembly. Ohmmeter indication shall not exceed 1 ohm.
   c. Lift junction box assembly from work bench (lift from one corner) until ohmmeter indicates more than 100k ohms. Distance of travel shall be between 0.125 and 0.250 inch.
   d. Lower junction box assembly to work bench surface. Ohmmeter indication shall not exceed 1 ohm.
   e. Lift junction box assembly cover (lift from edge with notch for conduit) while holding the junction box assembly securely on work bench until ohmmeter indicates more than 100k ohms. Distance of travel shall be between 0.125 and 0.250 inch.
   f. Disconnect ohmmeter test leads and install cover and cover screws.

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APPENDIX

TEST METHOD 2

FUNCTIONAL TEST

INSTALLATION TOOL

1. TEST APPARATUS:

   a. Wire - Belden 8053, or equivalent.
   b. Plywood and sheetrock, 1/2 inch or greater thickness.
   c. Spring scale.

2. TEST PROCEDURE:

   a. Assemble a spool of Belden 8053 wire to the installation tool.
   b. Install staples in the installation tool.
   c. Adjust tension on installation tool to maximum or as necessary to obtain penetration of approximately 1/4 inch in sheetrock.
   d. Drive 5 staples into the sheetrock, paying out wire between successive staples.
   e. Pay out another 2 or 3 inches and cut the wire.
   f. Look at the wire at the stapled points. The wire shall not be cut.
   g. Using pliers or other tool, grasp the wire and pull along its length with gradually increasing force until the wire breaks. The wire shall break without pulling out the staples.

      NOTE: If the wire pulls away from the staples adjust tension as necessary until steps d. through g. are successfully accomplished 2 times in a row.

   h. Substitute a piece of plywood for the sheetrock and perform steps d. through g.
   i. Attach spring scale to the free end of the wire after it is threaded through the wire guides on the installation tool.
   j. Hold the installation tool still and pull on the spring scale until the reel turns. Apply pull parallel to the base of the installation tool. Pull required to turn the reel shall be not less than 1 pound nor greater than 4 pounds.
   k. After completion of test, remove wire spool and staples and adjust tension to minimum.

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APPENDIX

TEST METHOD 3

WIRE STRENGTH-BREAK TEST

1. TEST APPARATUS:

   a. Spring scale.
   b. Stationary anchor.

2. TEST PROCEDURE:

   a. Cut a piece of 26-gauge grid wire 56 inches long (minimum) for use as a test specimen.
   b. Connect test wire between anchor and spring scale.

   NOTE: Ensure that there are no kinks in the wire before subjecting the wire to break strength testing.

   c. While observing the scale, apply a constant, steadily increasing load parallel to the length of the wire until the test wire breaks. The test wire shall break at 7.0 to 9.0 pound tension.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-52861C	2. DOCUMENT DATE (YYMMDD) 92/05/28

3. DOCUMENT TITLE
Sensor, Grid Wire, DT-545/FSS-9(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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